

HAT-P-36b

R-1699

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_200229 · created 2026-08-02T14:02:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458908.7743 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.044
Transit depth (Rp/Rs)^2	2.48 +/- 1.39 [%]
Orbital Inclination (inc)	82.59 +/- 2.87
Airmass coefficient 1 (a1)	0.63 +/- 0.28
Airmass coefficient 2 (a2)	0.247 +/- 0.078
Scatter in the residuals of the lightcurve fit is	39.1 %
Best Comparison Star	#1 - [510, 46]
Optimal Aperture	4.64
Optimal Annulus	8.82
Transit Duration (day)	0.084 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.157 $\pm$ 0.044 vs 0.126 (deviation 0.31 σ)
Duration fit vs published	0.084 d vs 0.0925 d (deviation 0.25 σ)
Mid-time O-C	8.4 min
Depth SNR	1.6
Residual scatter	39.1 %

Light curve

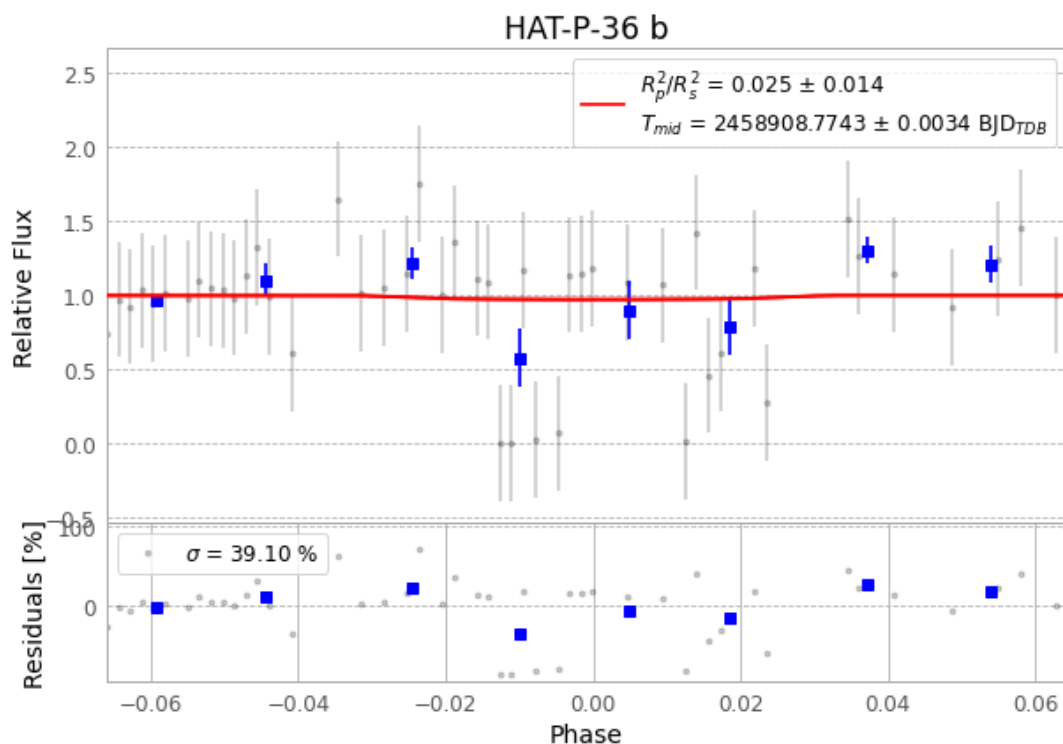


Figure 1 — FinalLightCurve_HAT-P-36 b_2020-02-28.png (SHA-256 511d5aa23ccc5da2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [401, 174], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 12946a626808d0f2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-36200229042112.FITS → HATP-36200229083312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-200229.log (7648afc4a7e9...), FinalLightCurve_HAT-P-36 b_2020-02-28.png (511d5aa23ccc...), FinalLightCurve_HAT-P-36 b_2020-02-28.csv (1e1e8ec8cad3...)

Compute resources

Wall time 220.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 14:02Z.